1. Math Foundations for ML

Priority	Topic
A	Vectors and matrices
A	Matrix multiplication
A	Dot product
A	Norms
A	Eigenvalues and eigenvectors
A	Linear transformations
A	Rank, determinant, inverse
A	Projections
A	Systems of linear equations
A	Derivatives
A	Partial derivatives
A	Chain rule
A	Gradients
A	Jacobian
A	Hessian basics
A	Probability basics
A	Conditional probability
A	Bayes theorem
A	Random variables
A	Expectation
A	Variance
A	Common distributions
A	Gaussian distribution

Priority	Topic
A	Bernoulli distribution
A	Binomial distribution
A	Multinomial distribution
A	Maximum likelihood estimation
B	Covariance matrix
B	Multivariate Gaussian
B	Lagrange multipliers
B	KL divergence
B	Entropy
B	Cross entropy
B	Information gain
C	Measure theory basics
C	Variational inference math
C	Bayesian nonparametrics

Minimum math for ML:

Linear algebra + calculus + probability + statistics + optimization

---

2. Statistics for ML

Priority	Topic
A	Mean, median, mode
A	Variance, standard deviation
A	Correlation
A	Covariance
A	Sampling

Priority	Topic
A	Bias and variance
A	Hypothesis testing
A	p-value
A	Confidence interval
A	Central limit theorem
A	Z-test, t-test
A	Chi-square test
A	A/B testing
B	ANOVA
B	Bootstrap
B	Bayesian statistics basics
B	Maximum likelihood estimation
B	Maximum a posteriori estimation
C	Markov Chain Monte Carlo
C	Causal inference
C	Do-calculus

For ML interviews, must know:

Bias-variance tradeoff

Overfitting vs underfitting

p-value

confidence interval

correlation vs causation

A/B testing

---

3. Python and ML Programming

Priority	Topic
A	Python basics
A	NumPy
A	Pandas
A	Matplotlib
A	Scikit-learn
A	Jupyter notebooks
A	Data loading
A	Data cleaning
A	Vectorization
A	Broadcasting
A	Train/test split
A	Pipelines
B	PyTorch
B	TensorFlow/Keras
B	Hugging Face Transformers
B	MLflow
B	FastAPI
B	Docker
C	CUDA programming basics
C	Triton kernels
C	Distributed training internals

Minimum tool stack:

Python + NumPy + Pandas + scikit-learn + PyTorch + Matplotlib

---

4. Data Preprocessing

Priority	Topic
A	Handling missing values
A	Handling outliers
A	Encoding categorical variables
A	One-hot encoding
A	Label encoding
A	Feature scaling
A	Standardization
A	Normalization
A	Train/validation/test split
A	Data leakage
A	Imbalanced datasets
A	SMOTE basics
A	Feature selection
A	Feature engineering
B	PCA preprocessing
B	Target encoding
B	Time-based split
B	Cross-validation split
B	Text preprocessing
B	Image preprocessing
C	Data versioning
C	Feature stores
C	Synthetic data generation

Most important warning:

Never let validation/test data influence training preprocessing.

That is data leakage.

---

5. Supervised Learning Algorithms

Regression

Priority	Algorithm
A	Linear Regression
A	Polynomial Regression
A	Ridge Regression
A	Lasso Regression
A	Elastic Net
A	Logistic Regression
B	Robust Regression
B	Quantile Regression
B	Bayesian Linear Regression
C	Gaussian Process Regression

Classification

Priority	Algorithm
A	Logistic Regression
A	K-Nearest Neighbors
A	Naive Bayes
A	Decision Tree
A	Random Forest
A	Support Vector Machine

Priority	Algorithm
A	Gradient Boosting
A	XGBoost
A	LightGBM
A	CatBoost
B	Extra Trees
B	AdaBoost
B	Linear Discriminant Analysis
B	Quadratic Discriminant Analysis
C	Gaussian Processes
C	Bayesian classifiers

Must study deeply:

Linear Regression

Logistic Regression

Decision Tree

Random Forest

SVM

Gradient Boosting

XGBoost/LightGBM

Naive Bayes

KNN

---

6. Unsupervised Learning

Priority	Algorithm / Topic
A	K-Means Clustering
A	Hierarchical Clustering

Priority	Algorithm / Topic
A	DBSCAN
A	PCA
A	t-SNE intuition
A	UMAP intuition
A	Anomaly detection
A	Gaussian Mixture Models
B	Association Rule Mining
B	Apriori Algorithm
B	Matrix Factorization
B	Autoencoders
C	Self-organizing maps
C	Spectral clustering
C	Isolation Forest deeply
C	One-class SVM

Must know:

K-Means

DBSCAN

PCA

Anomaly detection

Dimensionality reduction

---

7. Model Evaluation

Classification Metrics

Priority	Metric
A	Accuracy
A	Precision
A	Recall
A	F1-score
A	Confusion matrix
A	ROC curve
A	AUC
A	PR curve
A	Log loss
B	Top-k accuracy
B	Matthews correlation coefficient
B	Cohen’s kappa
C	Calibration error

Regression Metrics

Priority	Metric
A	MAE
A	MSE
A	RMSE
A	R^2 score
A	Adjusted R^2
B	MAPE
B	Huber loss
C	Quantile loss

Validation

Priority	Topic
A	Train/test split
A	Validation set
A	Cross-validation
A	K-fold CV
A	Stratified CV
A	Overfitting
A	Underfitting
A	Bias-variance tradeoff
B	Nested CV
B	Time-series validation
C	Bootstrap validation

Must know for interviews:

Precision vs recall

ROC-AUC vs PR-AUC

Cross-validation

Data leakage

Bias-variance tradeoff

---

8. Optimization for ML

Priority	Topic
A	Loss functions
A	Gradient descent
A	Batch gradient descent
A	Stochastic gradient descent

Priority	Topic
A	Mini-batch gradient descent
A	Learning rate
A	Momentum
A	Adam optimizer
A	RMSProp
A	Regularization
A	L1 regularization
A	L2 regularization
B	Learning rate schedules
B	Weight decay
B	Early stopping
B	Gradient clipping
B	Convex vs non-convex optimization
C	Second-order optimization
C	Natural gradient
C	L-BFGS
C	Sharpness-aware minimization

Must know:

Gradient descent

SGD

Adam

learning rate

regularization

early stopping

---

9. Deep Learning Basics

Priority	Topic
A	Artificial neuron
A	Perceptron
A	Multilayer perceptron
A	Forward propagation
A	Backpropagation
A	Activation functions
A	ReLU
A	Sigmoid
A	Tanh
A	Softmax
A	Loss functions
A	Cross entropy
A	MSE loss
A	Optimizers
A	Dropout
A	Batch normalization
A	Layer normalization
A	Weight initialization
A	Vanishing gradients
A	Exploding gradients
B	Residual connections
B	Skip connections
B	Normalization variants

Priority	Topic
B	Mixed precision training
C	Neural tangent kernel
C	Lottery ticket hypothesis

Deep learning workflow usually means working with data, defining models, optimizing parameters, and saving trained models, as PyTorch's beginner workflow also describes.

---

10. Neural Network Architectures

Core Architectures

Priority	Architecture
A	Feedforward Neural Network
A	CNN
A	RNN
A	LSTM
A	GRU
A	Encoder-decoder
A	Attention
A	Transformer
B	Autoencoder
B	Variational Autoencoder
B	ResNet
B	U-Net
B	Vision Transformer
B	Graph Neural Network
C	Neural ODE

Priority	Architecture
C	Capsule Network
C	Mamba / State Space Models
C	Mixture of Experts

Must study deeply:

MLP

CNN

RNN/LSTM/GRU

Attention

Transformer

Autoencoder

ResNet

U-Net

---

11. Computer Vision

Priority	Topic / Algorithm
A	Image representation
A	Convolution
A	Filters and kernels
A	Pooling
A	CNN architecture
A	Image classification
A	Object detection
A	Image segmentation
A	Transfer learning
A	Data augmentation

Priority	Topic / Algorithm
A	ResNet
A	VGG intuition
A	Batch normalization
B	YOLO
B	Faster R-CNN
B	Mask R-CNN
B	U-Net
B	Vision Transformer
B	CLIP
C	Diffusion for vision
C	SAM-style segmentation
C	3D vision
C	NeRF
C	Multimodal vision-language models

Must know:

CNN

ResNet

Transfer learning

YOLO basics

U-Net

Vision Transformer

Image augmentation

---

12. Natural Language Processing

Priority	Topic / Algorithm
A	Text preprocessing
A	Tokenization
A	Stemming
A	Lemmatization
A	Stopword removal
A	Bag of Words
A	TF-IDF
A	Word embeddings
A	Word2Vec
A	GloVe
A	RNN for text
A	LSTM for text
A	Attention
A	Transformer
A	BERT
A	GPT-style models
B	Sequence-to-sequence models
B	Beam search
B	Named Entity Recognition
B	Text classification
B	Sentiment analysis
B	Question answering
B	Summarization

Priority	Topic / Algorithm
B	Machine translation
C	Retrieval-augmented generation
C	Instruction tuning
C	RLHF
C	Preference optimization
C	Tool-using agents

Must know:

Tokenization

TF-IDF

Word embeddings

Attention

Transformer

BERT

GPT

Fine-tuning

RAG

---

13. Transformers and LLMs

Priority	Topic
A	Tokenization
A	Embeddings
A	Positional encoding
A	Self-attention
A	Multi-head attention
A	Feed-forward block

Priority	Topic
A	Residual connection
A	Layer normalization
A	Encoder-only models
A	Decoder-only models
A	Encoder-decoder models
A	Pretraining
A	Fine-tuning
A	Prompting
A	Context window
A	Temperature
A	Top-k sampling
A	Top-p sampling
B	Instruction tuning
B	RLHF
B	DPO
B	LoRA
B	QLoRA
B	PEFT
B	Quantization
B	KV cache
B	FlashAttention idea
B	RAG
B	Vector databases

Priority	Topic
B	Embedding models
B	Function calling / tool use
C	Mixture of Experts
C	Speculative decoding
C	Long-context methods
C	Multimodal LLMs
C	Agentic workflows
C	Post-training pipelines

Hugging Face's current learning tracks include LLMs, agents, context engineering, diffusion, deep RL, audio, vision, and other modern AI topics, so this section is now important for practical AI roles.

Must study deeply:

Transformer architecture

self-attention

tokenization

pretraining vs fine-tuning

RAG

LoRA/QLoRA

inference parameters

evaluation

---

14. Retrieval-Augmented Generation

Priority	Topic
A	Embeddings
A	Vector search

Priority	Topic
A	Semantic search
A	Chunking
A	Retrieval
A	Reranking
A	Context construction
A	Generation
A	Hallucination reduction
A	Evaluation
B	Hybrid search
B	BM25
B	Dense retrieval
B	Cross-encoder reranker
B	Query rewriting
B	Metadata filtering
B	Multi-hop RAG
C	Graph RAG
C	Agentic RAG
C	Fine-tuned retrievers

Must know:

chunking

embeddings

vector DB

retriever

reranker

context window

faithfulness

hallucination

---

15. Generative AI

Priority	Topic
A	Language models
A	Text generation
A	Image generation basics
A	Autoencoders
A	Variational Autoencoders
A	GANs
A	Diffusion models
A	Stable Diffusion basics
B	Conditional generation
B	Text-to-image models
B	Image-to-image models
B	ControlNet idea
B	CLIP
B	Multimodal generation
C	Flow matching
C	Video generation
C	3D generation
C	World models

Must know:

GAN

VAE

Diffusion

Transformer-based generation

CLIP

text-to-image basics

---

16. Reinforcement Learning

Priority	Topic / Algorithm
A	Agent
A	Environment
A	State
A	Action
A	Reward
A	Policy
A	Value function
A	Q-value
A	Markov Decision Process
A	Exploration vs exploitation
A	Bellman equation
A	Q-learning
A	SARSA
B	Deep Q-Network
B	Policy gradient
B	Actor-critic
B	Advantage function
B	PPO

Priority	Topic / Algorithm
B	A2C/A3C
C	Model-based RL
C	Multi-agent RL
C	Offline RL
C	RLHF
C	MuZero-style methods

For placements, RL is usually lower priority unless applying for research/robotics/AI roles.

Minimum RL:

MDP

policy

value function

Q-learning

DQN

policy gradient

PPO basics

---

17. Time Series ML

Priority	Topic / Algorithm
A	Time series data
A	Trend
A	Seasonality
A	Stationarity
A	Moving average
A	Exponential smoothing

Priority	Topic / Algorithm
A	Train/test split for time series
A	Forecasting metrics
B	ARIMA
B	SARIMA
B	Prophet
B	LSTM for time series
B	Temporal CNN
B	Transformer for time series
C	State space models
C	Kalman filter
C	Causal forecasting

Must know:

time-based split

leakage

ARIMA basics

LSTM forecasting

forecasting metrics

---

18. Recommendation Systems

Priority	Topic / Algorithm
A	Popularity-based recommendation
A	Content-based filtering
A	Collaborative filtering
A	User-user similarity

Priority	Topic / Algorithm
A	Item-item similarity
A	Matrix factorization
A	Embeddings
B	Neural collaborative filtering
B	Two-tower models
B	Ranking models
B	Cold start problem
B	Evaluation metrics
C	Session-based recommendation
C	Graph-based recommendation
C	Bandit-based recommendation

Must know:

collaborative filtering

content-based filtering

matrix factorization

cold start

ranking metrics

---

19. Graph Machine Learning

Priority	Topic
B	Graph representation
B	Node classification
B	Link prediction
B	Graph embeddings

Priority	Topic
B	DeepWalk
B	Node2Vec
B	Graph Neural Networks
B	GCN
B	GraphSAGE
B	GAT
C	Heterogeneous graphs
C	Knowledge graphs
C	Graph transformers
C	Geometric deep learning

Study this after normal DL unless specifically targeting graph ML.

---

20. MLOps and Deployment

Priority	Topic
A	ML project lifecycle
A	Experiment tracking
A	Model versioning
A	Data versioning
A	Model registry
A	Model serving
A	REST API deployment
A	Batch inference
A	Online inference
A	Monitoring

Priority	Topic
A	Drift detection
A	Logging
A	Reproducibility
B	Docker
B	CI/CD for ML
B	MLflow
B	DVC
B	FastAPI
B	Kubernetes basics
B	Feature store
B	A/B testing models
B	Shadow deployment
C	Kubeflow
C	Ray
C	Distributed serving
C	Model compression pipeline

Must know for ML engineer roles:

train model

save model

serve model

monitor model

handle drift

retrain pipeline

---

21. Model Compression and Efficient AI

Priority	Topic
B	Quantization
B	Pruning
B	Knowledge distillation
B	ONNX
B	TensorRT basics
B	Batch inference
B	Caching
B	Latency vs throughput
C	Low-rank adaptation
C	Sparse models
C	Edge deployment
C	Custom kernels

Important for practical AI products:

quantization

distillation

ONNX

latency optimization

batching

---

22. Responsible AI, Security, and Evaluation

Priority	Topic
A	Bias in data
A	Fairness
A	Explainability

Priority	Topic
A	Interpretability
A	Privacy
A	Data leakage
A	Model robustness
A	Adversarial examples
A	Hallucination
A	Evaluation benchmarks
B	SHAP
B	LIME
B	Model cards
B	Dataset cards
B	Red teaming
B	Prompt injection
B	Jailbreaks
B	PII handling
C	Formal verification
C	Differential privacy
C	Federated learning
C	AI governance

For LLM apps, must know:

hallucination

prompt injection

privacy

evaluation

red teaming

guardrails

---

23. Data Engineering Basics for ML

Priority	Topic
A	CSV/JSON/Parquet
A	SQL
A	Joins
A	Aggregations
A	ETL
A	Data pipelines
A	Batch processing
A	Data quality checks
B	Spark basics
B	Airflow basics
B	Kafka basics
B	Warehouses
B	Feature pipelines
C	Streaming ML
C	Lakehouse architecture
C	Distributed data processing internals

Must know:

SQL

data cleaning

ETL

feature pipelines

data validation

---

24. Complete Algorithm List: ML

Classical ML Algorithms

Linear Regression

Polynomial Regression

Ridge Regression

Lasso Regression

Elastic Net

Logistic Regression

Naive Bayes

K-Nearest Neighbors

Decision Tree

Random Forest

Extra Trees

Support Vector Machine

AdaBoost

Gradient Boosting

XGBoost

LightGBM

CatBoost

K-Means

K-Medoids

Hierarchical Clustering

DBSCAN

Gaussian Mixture Models

PCA

t-SNE

UMAP

Apriori

Isolation Forest

One-Class SVM

Matrix Factorization

Deep Learning Algorithms / Models

Perceptron

MLP

CNN

RNN

LSTM

GRU

Autoencoder

Variational Autoencoder

GAN

DCGAN

CycleGAN

Transformer

BERT

GPT

T5

ResNet

DenseNet

EfficientNet

U-Net

YOLO

Faster R-CNN

Mask R-CNN

Vision Transformer

CLIP

Diffusion Models

Stable Diffusion

Graph Neural Network

GCN

GraphSAGE

GAT

DQN

PPO

Actor-Critic

LLM / Modern AI Algorithms and Methods

Tokenization

BPE

WordPiece

SentencePiece

Self-Attention

Multi-Head Attention

Causal Language Modeling

Masked Language Modeling

Instruction Tuning

RLHF

DPO

LoRA

QLoRA

PEFT

Quantization

RAG

Dense Retrieval

BM25

Hybrid Search

Reranking

Function Calling

Tool Use

Agent Planning

Chain-of-Thought style reasoning

Evaluation and red teaming

---

25. Priority Study Order

Stage 1 — ML Foundations

1. Python
2. NumPy
3. Pandas
4. Matplotlib
5. Linear algebra
6. Probability
7. Statistics
8. Calculus basics
9. Optimization basics
10. Data preprocessing

Stage 2 — Classical ML

11. Linear Regression
12. Logistic Regression
13. KNN
14. Naive Bayes
15. Decision Tree

16. Random Forest
17. SVM
18. Gradient Boosting
19. XGBoost/LightGBM
20. K-Means
21. PCA
22. DBSCAN
23. Model evaluation
24. Cross-validation

Stage 3 — Deep Learning

25. Neural networks
26. Backpropagation
27. PyTorch
28. MLP
29. CNN
30. RNN/LSTM/GRU
31. Regularization
32. BatchNorm/LayerNorm
33. Optimizers
34. Transfer learning

Stage 4 — Modern AI

35. Attention
36. Transformer
37. BERT
38. GPT-style models
39. Fine-tuning
40. LoRA/QLoRA
41. RAG

42. Vector databases

43. LLM evaluation

44. Agents

45. Prompt engineering

Stage 5 — Specializations

46. Computer Vision

47. NLP

48. Recommendation Systems

49. Time Series

50. Reinforcement Learning

51. Graph ML

52. Generative AI

53. MLOps

54. Responsible AI

55. Efficient inference

---

26. Minimum Set for ML Placements

Study these first:

Python

NumPy

Pandas

Matplotlib

scikit-learn

linear algebra

probability

statistics

linear regression

logistic regression

decision trees

random forest

SVM

KNN

Naive Bayes

gradient boosting

XGBoost/LightGBM

K-Means

PCA

model evaluation

cross-validation

feature engineering

overfitting/underfitting

bias-variance tradeoff

---

27. Minimum Set for Deep Learning Roles

PyTorch

tensors

autograd

neural networks

backpropagation

activation functions

loss functions

optimizers

regularization

CNN

RNN

LSTM

GRU

attention

transformers

transfer learning

fine-tuning

GPU training basics

---

28. Minimum Set for Modern GenAI / LLM Roles

Transformers

tokenization

embeddings

self-attention

pretraining

fine-tuning

prompting

RAG

vector databases

reranking

LoRA

QLoRA

quantization

LLM evaluation

hallucination

prompt injection

agents

tool use

---

29. Minimum Set for ML Engineer Roles

classical ML

deep learning basics

data preprocessing

feature engineering

model evaluation

ML pipelines

experiment tracking

model deployment

FastAPI

Docker

MLflow

monitoring

drift detection

data versioning

model versioning

SQL

cloud basics

---

30. Recommended Problem / Project Count

Area	Minimum Practice
ML algorithms from scratch	8–10
Scikit-learn projects	5
Deep learning PyTorch projects	5
NLP projects	3
Computer vision projects	3
LLM/RAG projects	3

Area	Minimum Practice
MLOps/deployment projects	2
Kaggle competitions/notebooks	3–5
Research paper implementations	2–3

Best project set:

1. Tabular ML project
2. Image classification project
3. NLP text classification project
4. Recommendation system
5. Time-series forecasting
6. RAG chatbot over documents
7. ML model deployed with FastAPI + Docker

---

31. Your Best Study Focus

Since you are preparing for placements and also interested in AI systems, your highest ROI path is:

1. Classical ML fundamentals
2. PyTorch deep learning
3. Transformers and LLMs
4. RAG and AI agents
5. MLOps and deployment
6. One strong AI project

For a strong resume, build these:

1. End-to-end ML pipeline project
2. Deep learning project in PyTorch
3. RAG/LLM application with evaluation
4. Deployed ML API with Docker/FastAPI